

market that is sizeable enough to move the needle for the incumbent—it is too late. By that point, the smaller companies that took the time to master the new technology are much more nimble and experienced and will outcompete incumbents in what have become big growth markets.

If an incumbent misses enough of these growth opportunities, its offerings will become obsolete, its revenue will dwindle, its market capitalization will shrink, and it will become a dead-end investment. Often these are referred to as value-traps. As the innovative investor might expect, the fall of incumbents is happening at an accelerating rate, as is the rise of new winners. The disruption of incumbents can be quantified by how long the biggest companies stay in the S&P 500, or their average life span. The average life span for companies in the S&P 500 has fallen from 60 years in the 1960s to below 20 years of late.² This is clearly a sign that investors must not be complacent. One cannot assume that the companies succeeding today will continue to be the leading (and profitable) companies for decades to come.

Disruptive technologies are also being invented at an accelerating rate. The trend is one we have been witnessing for millennia. For example, between AD 900 and 1900, a new general purpose technology was invented roughly every 100 years, with notable examples including the steam engine, automobile, and electricity. In the twentieth century, a new general purpose technology came into existence every 15 years, with familiar examples like computers, the Internet, and biotechnology. In the twenty-first century, general purpose